CPO Experiment Plan

Synthetic experiments for the CPO project — implementation guide

CPO project plan

April 27, 2026

Abstract

This document specifies four synthetic experiments that together showcase the mechanism, the scaling behaviour, the operating regime, and the clusters-from-data extension of Cluster-Referenced Preference Optimization (CPO). Each experiment is small enough to run on a laptop in under one minute. For each experiment we state a **goal**, the **exact protocol** (data generation, architecture, training, logging, figure specification), and a **success criterion**: a concrete quantitative prediction that the experiment must either confirm or refute.

Common setup

The four experiments share the following skeleton. Implement once; reuse.

World. Prompts $x \in \{1, \dots, P\}$ with $P = 8$. Responses $y \in \{1, \dots, R\}$ per prompt with $R = 16$. Fixed true latent quality $q(x, y) \in [0, 1]$ drawn uniformly at world-construction time (seeded).

Annotator clusters. $K = 2$ clusters, each with a threshold τ_k and a label-noise rate ϵ_k. An annotator from cluster k who sees sample (x, y) returns label

$$s = \nVdash\{q(x, y) > \tau_k\} \oplus \mathrm{Bernoulli}(\epsilon_k),$$

where $\oplus$ denotes xor (random flip with probability ϵ_k). The cluster prior is $(\pi_A, 1 - \pi_A)$.

Dataset sampling. For experiments A–C: at each training step, sample a mini-batch directly from the population by drawing $(x_i, y_i) \sim \mathrm{Uniform}(P \times R)$, $c_i \sim \mathrm{Cat}(\pi_A, 1 - \pi_A)$, then labelling as above. For experiment D, sample a fixed offline dataset with annotator identity (see §4).

Policy. Tabular cluster-blind softmax: $\pi_\theta(y \mid x) = \mathrm{softmax}(\theta_{x,:})[y]$, $\theta \in \mathbb{R}^{P \times R}$, initialised to zero. Reference $\pi_{\mathrm{ref}}(y \mid x) = 1/R$ uniform.

Log-ratio reward. $r_i = \log \pi_\theta(y_i \mid x_i) - \log \pi_{\mathrm{ref}}(y_i \mid x_i) = \log \pi_\theta(y_i \mid x_i) + \log R$.

Loss. Unary prospect loss:

$$\mathcal{L}(\theta) = \mathbb{E}_i\big[\, 1 - \sigma(\beta\, m_i) \,\big], \quad m_i = \begin{cases} r_i - z_{c(i)} & s_i = D \\ z_{c(i)} - r_i & s_i = U \end{cases}$$

with $\beta = 1$ unless stated otherwise, and $\lambda_D = \lambda_U = 1$. The reference z is detached in the backward pass (stop-gradient).

Reference point (default variant). Log-ratio-mean over undesirable labels:

$$z_k \;\approx\; \mathbb{E}\big[r_i \mid c_i = k,\, s_i = U\big],$$

estimated by an EMA with rate $\rho = 0.1$:

$$z_k \;\leftarrow\; (1-\rho)\, z_k \;+\; \rho\, \tfrac{1}{|B_{k,U}|} \sum_{i \in B_{k,U}} r_i,$$

where $B_{k,U}$ is the subset of the mini-batch with cluster k and label U. For KTO, the same statistic is pooled over all clusters.

Optimiser. Adam, learning rate 0.15, batch size 128, 400 training steps.

Evaluation. Expected true quality under the current policy:

$$\mathbb{E}[q] \;=\; \tfrac{1}{P} \sum_x \sum_y \pi_\theta(y \mid x)\, q(x,y),$$

computed in closed form from the tabular logits. Record every 20 steps.

We pick $\mathbb{E}[q]$ because it measures **what we actually want** (latent response quality), is **independent of the labelling proxy** (so no method gets credit for overfitting to noisy labels), is **cluster-blind** (matching the cluster-blind policy), and is **closed-form computable** (no evaluation noise). It turns the question "did this method extract signal from corrupted labels?" into a single, comparable, well-calibrated number on $[0,1]$.

Seeds. All experiments use $n_{\text{seeds}} \in \{3,4,5\}$ independent seeds. Each seed regenerates the world (true q), the label stream, and the policy initialisation.

---

1 Experiment A — main comparison with gradient-weight diagnostic

Goal

Demonstrate the core mechanism: CPO outperforms KTO under base-rate heterogeneous annotators because *its per-cluster reference prevents the saturation of the sigmoid for the minority cluster.* The figure must not only show that CPO wins but *why*: per-cluster gradient weights $\sigma'(\beta m_i)$ should look qualitatively different between KTO and CPO.

Protocol

World parameters. $\pi_A = 0.9$. $\tau_A = 0.25$, $\tau_B = 0.75$. $\epsilon_A = \epsilon_B = 0.05$. $n_{\text{seeds}} = 4$.

Methods to compare.

- KTO on the full mixed dataset, single global z.
- CPO on the full mixed dataset, per-cluster z_A, z_B.
- *Oracle-Bob-only*: KTO trained only on the subset with $c_i = 1$ (Bob, strict). This baseline uses 1/10 of the data but clean labels; it is a useful ceiling.

Logging.

- Every 20 steps: evaluate $\mathbb{E}[q]$.
- *For the first seed only*, at every step, record the full per-sample gradient weight $\sigma'(\beta m_i) = \sigma(\beta m_i)\,(1-\sigma(\beta m_i))$ together with the cluster ID c_i.

Figure specification

A 1×3 panel figure, total size $\approx 15 \times 4.4$ inches.

Panel (a) — True-quality recovery. x-axis: training step. y-axis: $\mathbb{E}[q]$. Three curves: KTO (solid blue), CPO (solid orange), Oracle-Bob-only (dashed black). Shaded region = ±1 s.d. over seeds. Legend in lower-right.

Panel (b-KTO) — KTO gradient calibration. x-axis: training step. y-axis: per-cluster *mean* gradient weight $\overline{\sigma'(\beta m_i)}_k$, smoothed with a moving average of window 20. Two curves: Alice (green), Bob (red). Horizontal dotted line at 0.25 (the maximum of σ').

Panel (b-CPO) — CPO gradient calibration. Same as (b-KTO), but for CPO.

Success criterion

- **Quality gap:** $\mathbb{E}[q]_{\text{CPO}} - \mathbb{E}[q]_{\text{KTO}} \geq 0.15$ at the end of training.
- **Oracle beat:** $\mathbb{E}[q]_{\text{CPO}} > \mathbb{E}[q]_{\text{Oracle-Bob-only}}$ at the end of training.
- **Mechanism visible:** In panel (b-KTO), both clusters' gradient weights decay to near zero (<0.02) by the end of training. In panel (b-CPO), Bob's gradient weight remains elevated (mean over the last 25% of training >0.02) while Alice's decays normally.

---

2 Experiment B — the π_A sweep

Goal

Confirm a non-trivial *shape-level* prediction: the advantage $\Delta(\pi_A) = \mathbb{E}[q]_{\text{CPO}} - \mathbb{E}[q]_{\text{KTO}}$ is an inverted-U in π_A, peaking at moderate imbalance and vanishing at both extremes. Matching the predicted shape is much stronger evidence of mechanism than any single-setting comparison.

Protocol

World parameters. Thresholds and noise rates as in Experiment A. Sweep $\pi_A \in \{0.5, 0.7, 0.85, 0.95, 0.99\}$. $n_{\text{seeds}} = 3$.

Methods. KTO and CPO on the full dataset. No oracle in this experiment — the point is the shape of the relative advantage.

Figure specification

A 1×2 panel figure, total size $\approx 11.5 \times 4$ inches.

Panel (left) — Absolute quality. x-axis: π_A. y-axis: final $\mathbb{E}[q]$. Two curves with error bars: KTO (blue circles), CPO (orange squares). Error bars are ± 1 s.d. over seeds.

Panel (right) — Difference. x-axis: π_A. y-axis: $\Delta(\pi_A) = \mathbb{E}[q]_{\text{CPO}} - \mathbb{E}[q]_{\text{KTO}}$ with error bars (propagate error as $\text{s.e.}(\Delta) = \sqrt{\text{s.d.}^2_{\text{KTO}} + \text{s.d.}^2_{\text{CPO}}}/\sqrt{n_{\text{seeds}}}$). Horizontal black line at $\Delta = 0$.

Success criterion

- **Inverted-U shape:** Δ has a clear maximum at $\pi_A \in \{0.85, 0.95\}$ (not at the endpoints $\pi_A \in \{0.5, 0.99\}$).
- **Magnitude:** $\max_{\pi_A} \Delta \geq 0.08$.
- **Endpoints:** $\Delta(\pi_A = 0.5) < 0.04$ and $\Delta(\pi_A = 0.99) < 0.04$.

---

3 Experiment C — four ablations delimiting the regime

Goal

Characterise *when* CPO *helps and when it does not.* Four independent ablations, each isolating one factor. A good method is one whose failure modes are predictable.

Protocol — C1: reference variant

Setup. Same world as Experiment A ($\pi_A = 0.9$, base-rate heterogeneity). Train KTO and CPO under three reference variants:

1. $r \mid U$ **reference** (default): $z_k = \mathbb{E}[r_i \mid c = k, s = U]$.
2. r **reference**: $z_k = \mathbb{E}[r_i \mid c = k]$ (unconditional mean).
3. **KL reference** (as stated in the CPO paper): $z_k = \mathbb{E}_{x \in C_k}\big[\text{KL}(\pi_\theta(\cdot \mid x) \| \pi_{\text{ref}}(\cdot \mid x))\big]$.

Success criterion.

- **$r|U$ gives a gap:** Under the $r|U$ variant, CPO $\gg$ KTO (gap $\geq$0.15).
- **r and KL collapse:** Under the r and KL variants, CPO $\approx$ KTO (gap <0.03). *This is a substantive theoretical prediction:* with shared prompts, the KL reference is cluster-invariant because the prompt marginal is cluster-invariant.

Protocol — C2: noise-only heterogeneity

Setup. $\pi_A = 0.9$, but $\tau_A = \tau_B = 0.5$ (same threshold) and $\epsilon_A = 0.3$, $\epsilon_B = 0.02$ (different noise). The clusters differ only in label noise, not in base rate.

Success criterion.

- **No gap:** CPO $\approx$ KTO (gap <0.03). The CPO mechanism is base-rate calibration, not noise handling.

Protocol — C3: β sweep

Setup. Same world as Experiment A. Sweep $\beta \in \{0.3, 1.0, 3.0, 10.0\}$. $r|U$ reference.

Success criterion.

- **CPO advantage grows with β:** $\Delta(\beta = 10) > \Delta(\beta = 1) > \Delta(\beta = 0.3)$. The saturation-avoidance mechanism is more important at high β.

Protocol — C4: misspecified clusters

Setup. Same heterogeneous world as Experiment A, but the cluster ID passed to CPO is *drawn uniformly at random* (ignoring the true annotator type).

Success criterion.

- **No degradation:** CPO $\approx$ KTO (gap within seed noise, both $\pm$ bars overlap). CPO must not perform worse than KTO when clusters are meaningless.

Figure specification

A 1×4 panel figure, total size $\approx 17 \times 4.2$ inches.

- **Panel C1:** Grouped bar chart with 3 reference-variant groups, each with KTO (blue) and CPO (orange) bars.
- **Panel C2:** Two-bar chart: KTO, CPO.
- **Panel C3:** Line plot, x-axis β (log scale), y-axis final $\mathbb{E}[q]$, two curves (KTO, CPO) with error bars.
- **Panel C4:** Two-bar chart: KTO, CPO.

---

4 Experiment D — learned clusters

Goal

Answer Open Question 2 of the CPO manuscript: can clusters be *learned from data* rather than given? Using a simple annotator-signature-based K-means, we aim to recover the entire oracle benefit. We also demonstrate the bias-variance behaviour over K.

Protocol

Data generation. Static offline dataset with annotator identity.

Algorithm 1 Generate annotator-indexed dataset

```
Sample true quality q(x, y) ~ Uniform[0, 1], shape (P, R).
Draw N_ann = 120 annotators, each assigned type k_a ~ Bernoulli(1 − π*_A) with π*_A = 0.85.
For each annotator a = 1, ..., N_ann:
for each of n_per = 40 labels do
    Draw x ~ Uniform(P), y ~ Uniform(R).
    Label s = 1{q(x, y) > τ_{k_a}} ⊕ Bernoulli(0.05).
    Store tuple (x, y, s, a).
```

Clustering — 1-D signature. For each annotator a, compute the 1-D signature

$$\mathrm{sig}(a) = \frac{1}{|S_a|} \sum_{i \in S_a} s_i \;=\; \text{empirical } D\text{-rate of annotator } a,$$

where $S_a = \{i : a_i = a\}$. Run K-means on the $N_{\text{ann}} \times 1$ signature matrix; assign each sample's cluster as $c(i) = \text{cluster}(a_i)$.

Methods to compare at $K = 2$.

- KTO ($K = 1$, single global z).
- CPO with *random* cluster assignment per annotator (NMI ≈ 0 against truth).
- CPO with *learned* cluster assignment (1-D K-means).
- CPO with *oracle* cluster assignment (cheat, use true k_a).

K-sweep. For $K \in \{1, 2, 3, 5, 10\}$, compare:

- CPO with K-means on the 1-D signatures (*learned*).
- CPO with a stratified random partition of annotators into K equal-size groups (*random partition*). This isolates the effect of K on EMA-variance from any effect of clustering quality.

Training. Same CPO loop as in Experiments A–C, but samples are drawn uniformly at random from the pre-generated offline dataset rather than from the online label-generating process. Use the $r|U$ reference. $n_{\text{steps}} = 250$, batch size 128, learning rate 0.2.

Logging. Final $\mathbb{E}[q]$ for each method. For the clustering-quality comparison:

- Normalized mutual information $\mathrm{NMI}(c_{\text{pred}}, k_{\text{true}})$ (use `sklearn.metrics.normalized_mutual_info_score`).
- Purity: $\frac{1}{N_{\text{ann}}} \sum_k \max_{k^*} |\{a : c_{\text{pred}}(a) = k, k_a = k^*\}|$.

Figure specification

A 1×3 panel figure, total size $\approx 15 \times 4.2$ inches.

Panel (a) — Main bar chart. Four bars in order: KTO (blue), CPO random (grey), CPO learned (purple), CPO oracle (orange). y-axis: final $\mathbb{E}[q]$. Error bars = ± 1 s.d. Annotate each bar with its numerical value.

Panel (b) — Cluster recovery. Three groups (random, learned, oracle), each with two bars: NMI (dark) and purity (light). Error bars over seeds.

Panel (c) — K-sweep. x-axis: K (log-scaled). y-axis: final $\mathbb{E}[q]$. Two curves: CPO learned (purple squares), CPO random-partition (grey circles). Two reference horizontal lines: oracle $K = 2$ (orange dashed) and KTO $K = 1$ (blue dotted).

Success criteria

- **Learned ≈ oracle:** At $K = 2$, $|\mathbb{E}[q]_{\text{learned}} - \mathbb{E}[q]_{\text{oracle}}| < 0.02$.
- **Learned > random:** At $K = 2$, $\mathbb{E}[q]_{\text{learned}} - \mathbb{E}[q]_{\text{random}} \geq 0.05$.
- **Random ≈ KTO:** At $K = 2$, $|\mathbb{E}[q]_{\text{random}} - \mathbb{E}[q]_{\text{KTO}}| < 0.03$. Having "multiple clusters" is not enough; the partition must be meaningful.
- **Clustering recovery:** NMI of learned vs. true annotator type ≥ 0.9, purity ≥ 0.95.
- **Graceful over-clustering:** For $K \in \{3, 5, 10\}$, learned $\mathbb{E}[q]$ stays within 0.02 of the $K = 2$ value.
- **$K = 1$ collapse:** At $K = 1$, both curves match KTO exactly (no advantage from the CPO machinery when there is only one cluster).

---

5 Reporting deliverables

For each experiment, you should produce:

1. A self-contained Python script that runs the experiment end-to-end on a single machine in under 2 minutes.
2. The figure specified in the protocol, saved as PNG at 150 dpi.
3. A one-paragraph written interpretation addressing each success criterion: did it hold, did it not, and what does that tell us?
4. Cached intermediate results (e.g., a `.pkl` file) so the figure can be re-plotted without re-running the experiment.

6 Suggested order and timeline

- **Task 1.** Implement the common setup (world, policy, training loop, evaluation). Verify the loop reproduces KTO under $K = 1$ and zero heterogeneity.
- **Tasks 2–3.** Experiment A. The gradient-weight diagnostic is the most intellectually important part of the entire plan; take the time to stare at it.

- **Task 4.** Experiment B (fast; the sweep is cheap).
- **Tasks 5–6.** Experiment C. Four ablations; each one is its own mini-study.
- **Tasks 7–8.** Experiment D. Introduces scikit-learn for clustering.